Green Paper 06 — Planetary Boundaries as Ethical Constraint

When consequence returns to decision

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Abstract

Planetary boundaries describe Earth-system processes whose disruption can increase the risk of large-scale, abrupt, or irreversible environmental change. They are not walls around the planet, exact predictions of collapse, or moral verdicts on individual lives. They help identify a safe operating space for humanity — and the growing risks of moving beyond it.

This makes them ethically consequential. When actions can alter the conditions on which life depends, freedom cannot mean acting as though those conditions were infinitely replaceable. Yet planetary knowledge remains abstract unless it can meet actual places, decisions, institutions, and consequences without pretending that a global model is a local instruction.

This paper proposes translation without counterfeit precision. Earth-system science retains its scale and uncertainty. Situated knowledge retains its proximity and authority. Governance connects them through questions, observation, bounded action, and return. A planetary boundary becomes directly relevant when consequence is allowed to return to decision.

It also makes an ethical claim that cannot be derived from an aggregate risk model alone: every species carries value in itself. No species becomes expendable because a larger indicator remains within range, because another species performs a similar function, or because its disappearance can be made statistically small. This paper names that irreducible value sacredness. The word is an ethical orientation, not a demand for shared religion.

> The threshold that says: this changes you.

1. Boundaries are not verdicts

The planetary boundaries framework identifies nine critical processes that regulate the stability and resilience of the Earth system. The 2025 Planetary Health Check assesses seven of the nine boundaries as transgressed. This does not mean that seven switches have been thrown or that collapse will occur on a known date. A boundary is not necessarily a tipping point. It marks a level beyond which systemic risk rises and the safe operating space becomes less secure.

That distinction matters. Alarm without precision can immobilise. Precision without consequence can normalise danger.

The framework does not tell a municipality what to put in its budget, a farmer what to plant next season, or a neighbourhood how to share water. Nor should a global control variable be converted casually into a local quota. Its ethical force is different: it challenges decisions made as though ecological conditions were external, substitutable, or someone else's concern.

Limits are informational facts before they become rules. Crossing them does not make a person or society evil. It makes the conditions of action more fragile and the responsibility to perceive consequence more difficult to evade.

Yet information about systemic risk is not the whole of ecological ethics. The biosphere-integrity boundary necessarily works through selected variables and aggregated patterns. This makes planetary change legible, but it can also leave a dangerous remainder. A framework may show how much functional or genetic erosion destabilises the Earth system; it cannot, by that calculation alone, tell us what one unrepeatable species means or make its extinction ethically acceptable.

Quantification can tell us that a living relation is being lost. It cannot contain everything that is lost.

2. No species is collateral

> Every being has a right to refuse.

This paper takes a position: every species has intrinsic and unexchangeable value. Each is a singular expression of life, formed through evolutionary time and woven into relations that no inventory fully describes. A species is not only a service, a resource, a genetic library, a charismatic image, or a contribution to systemic resilience. It is itself.

The language of a sixth mass extinction names a scientific pattern of accelerating loss. It should not become a period label under which extinction is made to feel inevitable, normal, or historically complete before it has happened. Human societies should not knowingly drive a species from Earth, nor accept preventable extinction caused or accelerated by human action as the ordinary price of development.

This is an absolute orientation, but not a romantic description of ecology. Living systems include predation, disturbance, competition, migration, death, succession, and extinction. Habitats change. Species move. Human beings also alter landscapes simply by living. The claim is not that nature is static or harmonious, or that every organism can be preserved everywhere forever. It is that irreversible human-driven erasure must never be treated as an acceptable residual beneath an otherwise favourable aggregate.

This changes the grammar of Gold Before Bloom. Gold is not the quantity of biodiversity a field must accumulate before it earns a positive reading. Gold names what cannot rightfully be traded away. Where credible observation indicates that an action may destroy irreplaceable habitat, sever a necessary corridor, or contribute to the disappearance of a threatened species, an aggregate Green cannot cancel the protection claim. The signal requires attention, appropriate expertise, precaution, and, where necessary, pause.

3. Constraint changes what freedom means

Freedom inside a living system is not freedom from consequence. It is the capacity to act while remaining answerable to the conditions that make action possible.

Some consequences arrive quickly and nearby. Others are delayed, displaced, cumulative, or difficult to reverse. Carbon released here alters a shared atmosphere. Nutrient flows from many fields can return as oxygen-depleted water elsewhere. The loss of habitat may appear incremental until ecological relations can no longer reproduce themselves. The people and places receiving the consequences may be distant from those authorising or benefiting from the action.

This time- and distance-asymmetry changes the ethical field. Ordinary political and administrative systems are often organised around annual budgets, electoral terms, departmental mandates, and evidence that can be attributed within a manageable reporting period. Earth-system processes do not observe those boundaries.

Ethical constraint therefore does not mean that a scientific model governs society. It means that governance must become capable of remembering what its ordinary time horizons and jurisdictions tend to lose.

4. From planet to place without pretending equivalence

Planetary boundaries become directly relevant through a chain of translation, not by shrinking the planetary diagram until it appears local.

1. **Earth-system science names the larger relation.** It describes control variables, uncertainty, interactions, and zones of increasing risk at the scale for which the knowledge has standing.
2. **The boundary becomes an orientation question.** What larger living process might this decision affect? What does current science support, and what remains uncertain?
3. **A place encounters the question through an actual decision.** The inquiry begins with soil, water, work, food, habitat, infrastructure, procurement, finance, or another material relation — not with a demand to reproduce the global model.
4. **Different forms of knowledge meet without becoming interchangeable.** Situated observation may reveal what a remote

dataset cannot see. Scientific expertise may disclose relations that cannot be inferred locally. Administrative knowledge may clarify where authority and resources actually sit.
5. **The standing of each claim remains visible.** What is directly observed? What is measured? What is inferred? At what scale? By whom? What is still unknown?
6. **A bounded action is chosen.** Responsibility, scope, conditions for pause, and the time of return are named.
7. **Consequence answers back.** The next observation may confirm the action, complicate it, or require the model and decision to change.

This is relevance without false equivalence. The place does not contain the planet. The planetary model does not contain the place.

5. Bioregions: where the larger relation becomes inhabitable

Administrative territories are necessary for law, budgets, and public responsibility, but life does not organise itself primarily through municipal or national borders. Water crosses them. So do migratory routes, pollinators, fire regimes, seed dispersal, coastlines, soils, seasonal movements, and the effects of upstream action.

A bioregional reading asks how a particular place participates in these larger living relations. It does not replace local knowledge with another map. Nor does it declare a bioregion from a distance and require inhabitants to conform to it. It opens a conversation between mapped ecology and lived territory:

- Which watershed, aquifer, habitat network, coast, forest, grassland, or urban ecology is this place part of?
- Which species live, breed, feed, migrate, shelter, or pass through here?
- Which relations are visible only across seasons or across several places?
- Where have corridors been narrowed or severed?
- What names, memories, observations, and practices already exist here?
- Which administrative decisions are acting on one fragment of a relation that must remain connected?

Bioregional knowledge can correct a local frame that has become too narrow. Situated knowledge can correct a bioregional map whose categories fail to describe the living territory. The movement must remain reciprocal.

The purpose is not to assign each place responsibility for an entire planetary boundary. It is to make participation in larger relations perceptible enough that an actual decision can respond.

6. Citizen science: attention that can travel

Citizen science can help make an abstract planetary condition directly present without reducing local observers to suppliers of raw data. A

recurring record of first and last sightings, abundance, breeding, flowering, insect presence, water condition, soil cover, habitat use, seasonal timing, disturbance, or absence may reveal changes long before they appear in a distant institutional cycle.

The observation is full knowledge at its own scale: full but not final, situated, fallible, and correctable. A fisher, farmer, child, gardener, herder, birdwatcher, neighbourhood, or Indigenous knowledge holder may perceive a relation that an aggregate dataset cannot see. A specialist or larger dataset may disclose what local observation alone cannot establish. Neither should be made raw material for the other.

The 13×13 citizen-science grammar can assist by asking across ecological, bodily, relational, technological, and governance dimensions what is changing, how consequence is felt, where power sits, and how correction becomes possible. It is an inquiry structure, not a requirement and not a universal species inventory. Data is offered, not extracted; some knowledge may properly remain local, protected, seasonal, relational, or unrecorded.

Epistemic standing must remain explicit. The disappearance of a species from one observed place is serious local knowledge, but it is not by itself proof of global extinction. A global assessment cannot by itself explain why the species vanished here. The responsible movement connects observations, compares periods and places, invites relevant ecological and taxonomic knowledge, and keeps uncertainty visible.

This creates a possible path:

attention → observation → shared memory → bioregional relation → scientific inquiry → bounded decision → consequence → correction

The value is not only that citizens produce more data. It is that people and places retain standing in the interpretation of the living changes they encounter.

7. A dashboard is not the world

The Penguin Dashboard offers one small governance surface for this translation. It keeps three streams distinct:

- **Land / Life**
- **Steward Viability**
- **Governance / Commons**

Improvement in one stream cannot erase harm in another. Ecological gain cannot compensate for exhausted stewards. Human wellbeing cannot erase ecological degradation. Administrative order cannot erase either. Green, yellow, red, and unknown are documented readings, not properties of reality, and they are not combined into a total score.

Planetary boundaries remain a horizon of constraint and relation rather than a fourth Dashboard stream. The Dashboard does not invent local planetary

limits. It helps a field notice when a situated question touches a larger process and when additional science, data, experience, or expertise is needed.

A Land / Life reading must also be able to carry a **species or habitat protection flag**. The flag is not another colour and cannot be averaged into a total score. It records that a potentially irreversible relation requires separate attention. A field should not describe itself as ecologically viable while credible evidence of serious harm to a threatened species or irreplaceable habitat remains unanswered.

AI may assist this movement between pixel and planet if it remains in right relationship. It can help organise observations, retrieve relevant knowledge, compare scales, expose uncertainty, and formulate questions. It must not manufacture local authority, determine field truth, conceal the provenance of claims, or convert uncertainty into confident instruction.

A responsible assistance might say:

> Here is what these observations appear to support. Here are larger relations that may be relevant. Here is what cannot be inferred from the available material. Here are people, knowledge communities, datasets, or forms of expertise that may help.

The system assists the reading. It does not own the field. The Dashboard succeeds partly when it reveals the boundary of its own knowledge.

8. Uneven consequence without fixed victim identities

Planetary disruption is not produced, experienced, or escaped evenly. Exposure, benefit, authority, adaptive capacity, and responsibility are distributed through histories of land, wealth, extraction, labour, infrastructure, and political power.

This can be described without assigning permanent identities of strength or vulnerability to whole groups. A person may hold authority in one relation and be excluded in another. A community may possess deep ecological knowledge while lacking formal standing in the institution deciding its future. An organisation may be financially secure while dependent on exhausted people or damaged places whose signals do not enter its accounts.

The relevant questions are structural:

- Who receives the benefit, and who receives the consequence?
- Who can alter the decision, and who can only adapt to it?
- Whose observations have standing?
- Who has time, money, mobility, legal protection, or ecological room to absorb disruption?
- Which burdens have been displaced across distance, generations, species, or institutions?

- What repair, restraint, or differentiated obligation follows from that distribution?

Justice under constraint is not achieved by giving everyone the same allowance after ecological space has already been occupied unevenly. Nor is it achieved by speaking about people only through deficiency. It requires honest accounting of relations and sufficient transfer of authority, capacity, and resources for consequence to change the decision.

9. A connected ecology of guardianship

Planetary knowledge reaches public life through more than one kind of node. It may move through Earth-system research, health checks, visualisation, public leadership, Indigenous and place-based knowledge, policy, finance, education, media, civic organisation, and local practice.

The independent Planetary Guardians initiative associated with Johan Rockström works across several of these interfaces. It elevates the planetary boundaries science through the Planetary Health Check, public communication, visual and sonic interpretation, and an international group spanning science, Indigenous leadership, politics, business, finance, activism, and conservation.

Planetary Guardians within Spiralweb has a separate origin and governance. Its emphasis is relational and situated: relationship before programme; local authority; field status distinct from protocol status; observation without forced standardisation; support without capture; and correction when consequence returns.

There is no affiliation to imply. The shared name, used by the two initiatives, does not make the initiatives one organisation. Yet their different routes illuminate a wider possibility. One ecology can help planetary science become publicly visible and institutionally present. Another can explore how living places speak back into coordination, learning, and decision. Neither route contains the whole.

The work is not to establish a single centre. It is to improve the passages between nodes while preserving provenance, difference, and the possibility of correction.

10. Sacredness without spiritual authority

> Every field has a rhythm to be honoured.

The poetic source of this paper uses the language of the sacred. That word carries religious, spiritual, cultural, political, and personal histories. It cannot be presumed to mean the same thing to everyone. Yet avoiding the word entirely would also conceal the ethical position from which this paper is written.

Here sacredness names the intrinsic, irreducible worth of living beings and relations: their refusal to be exhausted into utility, price, aggregate function, or human preference. The paper stands by that orientation. It does not claim to prove sacredness scientifically, translate it on behalf of others, or establish a spiritual authority around it.

We do not need a common theology to recognise a common interest in conditions within which people, planet, and life can thrive. People may understand those conditions through scientific, secular, religious, spiritual, Indigenous, artistic, philosophical, or other traditions. No one interpretation is a condition of participation, and no steward or system speaks for the sacred on another person's behalf.

The public commitment is nevertheless clear: no species is merely collateral. Some relations should not be exhausted merely because an actor has the temporary power to exhaust them. Boundaries can protect the possibility of continued life, difference, refusal, renewal, and return.

11. Governance that can be changed by what it learns

A boundary-respecting institution does more than add environmental indicators to an unchanged decision process. It builds the capacity to be altered by what it discovers.

This includes:

1. keeping global science, local observation, and political judgment distinct but connected;
2. recording provenance, scale, uncertainty, and unknowns rather than laundering them into one authoritative score;
3. maintaining precautionary buffers where consequences may be irreversible;
4. ensuring that ecological ambition is not financed through human exhaustion or displaced harm;
5. naming no more actions than a field can responsibly carry;
6. assigning responsibility and a time for returning to observed consequence;
7. enabling pause, refusal, revision, repair, and retreat;
8. preserving public memory beyond projects, budgets, and electoral terms;
9. allowing those affected by a decision — including situated knowledge holders — to change its course rather than merely comment on it;
10. treating credible threats to species, habitats, and ecological relations as protection claims that cannot be averaged away; and
11. correcting both the action and the framework when reality does not behave as expected.

The spiral is not repetition for its own sake. It is the return through which a model encounters what it omitted.

12. Where the planetary conversation stands

The planetary conversation has changed during the past decade. Climate and biodiversity were once commonly joined through the language of co-benefits: protect a forest and it may store carbon, shelter species, retain water, and support livelihoods at the same time. That remains true, but it is no longer an adequate description of the field.

The conversation has moved toward **interdependence**, **nexus**, and **joint implementation with visible trade-offs**. Climate change is a major driver of biodiversity loss. Damaged ecosystems weaken climate regulation and resilience. At the same time, an intervention presented as climate action can damage species, simplify ecosystems, displace people, or reorganise a living place around a distant carbon claim. A plantation is not an ancient forest. A carbon gain does not automatically constitute a biodiversity gain.

Several major frameworks now meet around this problem, but they do different work:

Framework	What it makes legible	What it cannot decide alone
Planetary Boundaries	Earth-system stability, functional integrity, and systemic risk	Which particular species or bioregional relation must be protected
IUCN Red List	Documented global extinction risk for assessed species	A species' intrinsic worth or the political action that should follow
CBD / Global Biodiversity Framework	Shared international goals for ecosystems, species, restoration, use, finance, and participation	How a concrete place should interpret and enact those goals
IPBES	Assessed knowledge about biodiversity, drivers, values, and the biodiversity–water–food–health–climate nexus	Local mandate, legitimate authority, or an automatic decision
UNFCCC / IPCC	Climate commitments and assessed climate science	The full ecological meaning of a climate intervention in a particular habitat

The 2021 IPBES–IPCC workshop helped make the mutual dependence of climate and biodiversity explicit. The 2022 Kunming–Montreal Global Biodiversity Framework connected biodiversity action with climate mitigation, adaptation, and ocean acidification. The 2024 IPBES Nexus Assessment considered biodiversity, water, food, health, and climate together. Across the wider family of biodiversity-related conventions, coordination is growing.

The development is real but incomplete. Treaties, ministries, funds, datasets, assessment cycles, and professional communities remain divided. Integration can also become another aggregation problem: several unlike harms may disappear inside a single positive account. The emerging discipline is therefore not simply to combine more indicators. It is to connect forms of knowledge while keeping their objects, scales, uncertainties, conflicts, and authorities visible.

For this paper, the resulting field can be read through four connected lenses:

1. **Planetary condition** — Is Earth-system stability being placed at increasing risk?
2. **Species and ecosystem condition** — Which forms of life and ecological relations face decline or extinction?
3. **Bioregional relation** — How does the particular place participate in watersheds, habitats, corridors, seasons, and larger living systems?
4. **Situated observation** — What is actually being noticed here, by whom, with what standing, and with what consequence for the next decision?

No lens replaces the others. Together they make it harder for a global abstraction to overlook a species and harder for a local decision to overlook the planet.

13. From planetary intention to rules in use

International goals do not act by themselves. They become real through budgets, permits, land practices, professional judgments, community agreements, monitoring routines, conflicts, refusals, and repair. In the language of the Bloomington School, formal commitments meet **rules in use**: the rules people actually follow, contest, adapt, and maintain in concrete action situations.

Vincent and Elinor Ostrom’s work remains central because it rejected the assumption that complex shared conditions must be governed from one centre. Polycentric governance begins from a practical limit: no single institution possesses enough knowledge, legitimacy, or operational capacity to see and govern the whole. Multiple centres can act with meaningful autonomy while remaining connected through nested arrangements, public law, monitoring, conflict resolution, and mutual accountability.

This does not mean that every local practice is ecologically sound or that global coordination is unnecessary. A community cannot solve upstream pollution, atmospheric change, migratory decline, international finance, or national law alone. Nor should a distant institution treat local observation as anecdote while retaining sole authority to define reality. The institutional question is how different centres can become mutually consequential without one absorbing the others.

The Institutional Analysis and Development tradition offers a modest but powerful set of questions: Who participates? What positions and actions are available? What information is present or absent? Which rules are actually

in use? How are benefits and burdens distributed? Who can change the rule? What happens after the decision?

Within Spiralweb, the **PG Ledger** grows from this constitutional problem. Its purpose is not to centralise truth or turn polycentric governance into ledger governance. It helps a field remember:

- what was observed;
- what was inferred;
- whose knowledge had standing;
- what remained uncertain or contested;
- which rule or decision was used;
- who held responsibility;
- who or what carried the consequence; and
- what later required correction.

The Ledger came early in the Spiralweb inquiry because memory is the hinge between action and learning. Without a contestable record, consequences can be displaced, warnings forgotten, responsibility fragmented, and the same harm renamed as a new beginning. With the wrong record, however, governance becomes surveillance or bureaucracy. The record must therefore remain proportionate, consent-aware, locally useful, and smaller than the life it serves.

A possible polycentric chain is:

situated observation → structured inquiry → assisted pattern recognition → shared and contestable memory → named judgment → action → consequence → correction

Citizen science can make changes visible. The 13×13 can help structure inquiry. AI can connect scales and recover patterns. The PG Ledger can hold memory and disagreement. Public institutions can provide law, resources, continuity, and democratic review. Local and bioregional centres can retain standing, refuse false categories, and adapt rules to actual conditions. None governs alone.

This is how rules in use may become genuinely co-creative with planetary intention. The world’s shared goals do not merely descend into places for implementation. Places can answer back, reveal what the global frame omitted, and participate in changing the next iteration.

Working propositions

1. Planetary boundaries have ethical force without being moral verdicts or exact predictions of collapse.
2. A global boundary should not be converted into a locally invented quota without scientifically warranted translation.
3. Every species has intrinsic and unexchangeable value; no favourable aggregate can make preventable human-driven extinction ethically acceptable.
4. Gold is not an ecological score. It names what must not be traded away.

5. Direct relevance begins with an actual decision and the living relations it may alter.
6. Bioregional reading connects situated places with watersheds, habitats, corridors, seasons, and other relations that exceed administrative borders.
7. Knowledge claims become more trustworthy when their provenance, scale, uncertainty, and limits remain visible.
8. Situated knowledge and Earth-system science can correct one another without being treated as interchangeable.
9. Citizen science becomes transformative when people retain epistemic standing, consent, and a role in interpretation rather than only supplying data.
10. No ecological improvement can compensate automatically for human depletion or governance failure.
11. Justice under constraint requires attention to the distribution of benefit, consequence, authority, and capacity rather than fixed identities of vulnerability.
12. AI is useful when it increases legibility and corrigibility without acquiring authority it has not been given.
13. A governance system is responsive only when consequence can alter its next action.
14. Uncertainty does not dissolve responsibility. It changes how carefully we proceed, what we observe, and whether we remain able to stop.

Questions for practice

- What actual decision are we trying to inform?
- Which planetary relation may be relevant, and what does the science genuinely support?
- What can be observed here that a planetary model cannot see?
- What larger relation cannot be inferred from local observation alone?
- Which species, habitats, corridors, or seasonal relations may be affected?
- Is a threatened species or irreplaceable habitat being hidden inside an aggregate reading?
- What bioregional relation does the administrative frame omit?
- Who already observes this place, and under what conditions may that knowledge be shared?
- Who benefits, who carries consequence, and who can change the decision?
- What remains unknown?
- What is the smallest responsible next action?
- Who holds it, when do we return, and what would make us pause?
- How will consequence reach the next decision?

Textual provenance and working lineage

The italic passages are drawn from Gaia Bloom Constitution (AbunDance, 2025), especially Wild Law & Ritual Boundaries. The poetic material is used as constitutional imagination, not as empirical evidence or spiritual doctrine.

The paper’s later formulation grows through the wider Spiralweb inquiry, including Moral Biology, Regulation: Body · Relation · Institution, Viability and the Conditions for Care, Attention as Ethics, Commons, Trust, and Local Stewardship, Gold Before Bloom, Boundaries Are Not Violence, Knowing From the Ground, the Gaia GoldBloom citizen-science and 13×13 materials, Rule of Life, The Correction Loop, Penguin Dashboard: Legibility as Governance, The Founding Field, and the developing Planetary Guardians field architecture.

Scientific grounding includes the planetary boundaries framework initiated by Rockström and colleagues, subsequent assessments led by Steffen, Richardson, and colleagues, and the annual Planetary Health Check. These sources establish the Earth-system analysis. They do not prescribe the situated governance architecture proposed here.

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Read with

- Green Paper 01 — Moral Biology
- Green Paper 02 — Regulation: Body · Relation · Institution

- Green Paper 03 — Viability and the Conditions for Care
- Green Paper 04 — Attention as Ethics
- Green Paper 05 — Commons, Trust, and Local Stewardship
- Green Paper 07 — The Civic Nervous System
- Green Paper 10 — Notes Toward Planetary Guardianship
- Green Paper 11 — Gold Before Bloom
- Green Paper 18 — Boundaries Are Not Violence
- Report 02 — The Correction Loop
- Report 04 — Penguin Dashboard: Legibility as Governance
- Knowing From the Ground
- Gaia GoldBloom Citizen Science — 13×13 Nodes
- Rule of Life

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